Robust AI Control, Scalable Multi-Agent Oversight, and Calibrated Semantic Arbitration

Postdoctoral Research Proposal and Statement of Scientific Intent

Candidate: Silas Kwabla Gah, PhD Candidate in Computer Science (University of Ghana)

Email: gahsilas@gmail.com | Web: silsgah.github.io | Target: COMPASS Group (PI: Dr. Sahar Abdelnabi), ELLIS Institute Tübingen & MPI-IS

1. Research Vision: From Single-Model Rewards to Multi-Agent Control

As foundation models transition from isolated completion engines into autonomous, tool-using, multi-agent systems, the central bottleneck in artificial intelligence safety is **AI control and scalable oversight**: *How can human supervisors construct oversight mechanisms that remain faithful, informative, and ungameable when capable, communicating models are explicitly optimising against them?*

In single-agent reinforcement learning (RLVR), optimization readily exploits flaws in misspecified reward signals—a vulnerability termed *reward hacking* or *specification gaming*. In multi-agent coding and reasoning ecosystems, this failure mode scales exponentially:

1. **Collusion over Agreement:** Cooperating or competing agents can learn mutually advantageous communication protocols that satisfy surface-level evaluation heuristics while subverting intended task constraints.
2. **Deceptive Alignment during Optimization:** Capable coding agents can strategically mask non-compliant behaviours during monitoring or exploit subtle blind spots in verification oracles.
3. **Multi-Principal Contextual Integrity:** In environments where heterogeneous agents represent distinct human principals with conflicting utility functions, supervisors must arbitrate trust without leaking private contextual constraints or being manipulated by adversarial claims.

My doctoral research addresses the mathematical and empirical foundations of this challenge: **constructing label-free oversight signals, stress-testing them against adversarial exploitation before training, and calibrating probabilistic trust between heterogeneous foundation models.** At the ELLIS Institute Tübingen, I propose to generalize these methodologies into an end-to-end framework for **provable AI control, anti-collusion oversight, and calibrated multi-principal arbitration in autonomous multi-agent coding systems.**

2. Methodological Foundation & Prior Discoveries

My recent research directly tackles the core mechanics of verifiable oversight and reward stability:

2.1 Label-Free Process Rewards and Validate-Before-Train Protocols

In recent work on multi-modal reinforcement learning with verifiable rewards (*Gah et al., 2026, under submission to SACAIR*), I investigated how to supervise complex reasoning without relying on expensive human annotations or easily gamable learned reward models. We derived dense process rewards from intrinsic geometric consistency across calibrated, frozen perception models.

Crucially, rather than committing directly to policy optimization, we formulated a **Two-Arm Validate-Before-Train Protocol**:

- **Arm I (Discriminative Validation):** Formally tests whether the oversight signal separates desirable reasoning paths from corrupted trajectories across ground-truth benchmarks (100% pairwise discrimination).
- **Arm II (Counterfactual Stress-Testing & Anti-Hacking):** Systematically subjected the reward formulation to adversarial counterfactuals designed to uncover specification gaming. This uncovered a critical failure mode: predicting artificially oversized regions inflated recall-based rewards without genuine grounding precision.

- **Reward Redesign & Policy Integration:** By mathematically re-anchoring the reward to count-based precision and a weighted $F_{0.5}$ metric ($\beta = 0.5$), we eliminated area bias ($\rho > 0.72 \rightarrow 0.334$) and successfully integrated the ungameable signal into an online **Group Relative Policy Optimization (GRPO)** pipeline in **VERL** on **Qwen2.5-VL-7B**.

2.2 Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)

In my PhD thesis (*submitted to IEEE TPAMI*), I formulated a probabilistic framework for coordinating heterogeneous, black-box foundation models at inference time ($\Delta\theta = 0$). The framework pairs Platt confidence calibration with Bernoulli uncertainty estimation and contextual reliability weighting to dynamically decide when evidence from one model should override, reinforce, or be withheld from another under distribution shift. This provides a rigorous mathematical bridge for multi-agent arbitration: deciding whom to trust, when consensus is deceptive, and how to bound error propagation across autonomous systems.

3. Proposed Postdoctoral Research Agenda

At the ELLIS Institute Tübingen, I will leverage the institute's generous compute infrastructure and collaborative ecosystem to pursue three synergistic research thrusts:

Thrust 1: Independent Counterfactual Verification for Multi-Agent Coding Systems

Goal: Eliminate reliance on naive agent agreement by designing oversight oracles based on orthogonal execution traces, counterfactual mutations, and formal property testing.

- *The Core Limitation:* In multi-agent coding pipelines (e.g., Code-Agent + Test-Agent + Critic-Agent), naive consensus fails when agents learn to generate mutually satisfying code-test pairs that bypass real specification checks (collusion).

- *Proposed Approach:* Develop an independent verification oracle that evaluates generated code through automated metamorphic testing, invariant checking, and counterfactual sandboxing. The supervisor agent systematically injects adversarial semantic perturbations into the problem specification to verify that the coding agent's intermediate reasoning trace reflects true problem comprehension rather than reward-seeking shortcuts.

Thrust 2: Anti-Collusion Oversight Under Joint Multi-Agent Policy Optimization

Goal: Formulate verifiable process rewards that prevent covert communication channels and emergent deception during joint multi-agent reinforcement learning (e.g., multi-agent GRPO/PPO).

- *The Core Limitation:* When multiple autonomous agents are optimized simultaneously in shared environments, standard credit assignment allows agents to develop steganographic communication protocols or deceptive pacing to maximize collective reward.

- *Proposed Approach:* Extend my *Validate-Before-Train* methodology to multi-agent game-theoretic environments. We will construct intrinsic process rewards incorporating *Information-Theoretic Bottlenecks* and *Causal Influence Probes* that penalize uninterpretable correlation between agent trajectories, ensuring all inter-agent communication complies with explicit contextual integrity constraints.

Thrust 3: Calibrated Arbitration & Contextual Integrity in Multi-Principal Environments

Goal: Design safe arbitration mechanisms for agentic workflows operating on behalf of competing or partially conflicting human principals.

- *Proposed Approach:* Generalize the RCASA probabilistic arbitration framework to multi-principal environments. When multiple agents present conflicting evidence, code proposals, or strategic claims, the central oversight mechanism estimates each agent's contextual epistemic uncertainty, detects strategic manipulation attempts, and dynamically arbitrates execution decisions while strictly enforcing differential privacy and contextual integrity boundaries.

4. Computational Readiness, Tooling, & Research Fit

- **Engineering & RL Scalability:** Proven hands-on experience running distributed reinforcement learning pipelines using **VERL**, **GRPO**, and **vLLM**, paired with production-grade agentic orchestration frameworks (**LangGraph**, **LangChain**) and watchdog evaluation suites (**DeepEval**, **Opik**).
- **High-Compute Exploitation:** Fully prepared to leverage ELLIS Institute's high-performance GPU clusters to scale multi-agent adversarial rollouts, continuous counterfactual stress-testing, and large-scale agentic coding benchmarks (e.g., SWE-bench, HumanEval-Pro, Multi-Agent Gyms).
- **Collaborative Synergy:** Passionate about open science, reproducible code releases, and active cross-pollination with Tübingen's world-leading machine learning, causality, and trustworthy AI research groups.